

Data Analytics Life Cycle

HR Attrition Analytics Dashboard (Power BI)

Document control	
Project	HR Attrition Analytics Dashboard
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Related documents	Business Requirements Document (BR-01 to BR-10)

1. Overview

This document describes how the HR Attrition Analytics project follows a full analytics life cycle, from understanding the business problem to publishing and improving the solution. It shows what is done at each stage, which tools are used, what is produced, and how to tell that the stage is complete.

The project uses the IBM HR Employee Attrition dataset (1,470 employees, 35 fields, 16.1% attrition). The data is synthetic, so the project demonstrates the method rather than real-world conclusions.

2. Life cycle at a glance

#	Stage	Key deliverable	Repository location
1	Ask / business understanding	Business Requirements Document	/docs
2	Data acquisition	Data source note	/data/raw
3	Data profiling and cleaning	Cleaned dataset, cleaning log, data dictionary	/data, /docs
4	Data modelling	Star schema and model diagram	/model
5	Exploratory analysis	Findings notes	/insights
6	Measures and validation	Measure list and validation log	/model
7	Visualisation and report design	Power BI report (.pbix)	/report
8	Insights and communication	README insights and summary	/insights, README
9	Review and acceptance	Traceability matrix and review log	/docs
10	Deploy, monitor, and iterate	Publishing notes and roadmap	README

3. Stage details

Stage 1: Ask / business understanding

Objective	Agree the business problem, objectives, stakeholders, and scope before touching any data.
Key activities	• Frame the problem: 237 of 1,470 employees have left (16.1%) and the drivers are unclear. • Identify simulated stakeholders and their interests. • Set in-scope and out-of-scope items and prioritise requirements with MoSCoW. • Write the BRD (BR-01 to BR-10) with acceptance criteria.
Tools	Word, Excel
Deliverables	Business Requirements Document.
Exit criteria	Requirements and scope agreed by the (simulated) sponsor.

Stage 2: Data acquisition

Objective	Obtain the data and document where it came from and what it contains.
Key activities	• Download the IBM HR Employee Attrition CSV from Kaggle. • Record the source, licence, and download date. • Review the 35 fields against the dataset description. • Note that the data is synthetic and has no date field.
Tools	Kaggle, GitHub
Deliverables	Data source note; unaltered raw file.
Exit criteria	Raw file stored unchanged and provenance documented.

Stage 3: Data profiling and cleaning

Objective	Confirm the data is fit for purpose and prepare it for modelling.
Key activities	• Confirm 1,470 rows and 35 columns, and check for null values (none found). • Confirm EmployeeNumber is unique. • Remove the constant columns EmployeeCount, StandardHours, and Over18. • Derive a 1/0 IsAttrition field from Attrition. • Replace numeric rating scales with descriptive labels. • Check ranges for outliers (for example age 18–60, monthly income 1,009–19,999). • Record every change in a cleaning log.
Tools	Power Query, Python (pandas), SQL
Deliverables	Cleaned dataset, cleaning log, data dictionary.
Exit criteria	Row count still 1,470 and every change is logged and explained.

Stage 4: Data modelling

Objective	Structure the data as a star schema so measures and slicers behave predictably.
Key activities	• Design FactEmployee at employee grain (1,470 rows). • Create DimDepartmentRole keyed on the department and job role combination (11 combinations, because some roles span departments). • Create DimEducationField, DimEmployeeProfile, band tables (age, salary), and DimRating. • Define one-to-many relationships with single-direction filtering.
Tools	Power Query, Power BI model view
Deliverables	Model diagram and table definitions.
Exit criteria	Fact table still has 1,470 rows and there are no orphan keys.

Stage 5: Exploratory analysis

Objective	Find where attrition concentrates and test candidate drivers before designing visuals.
Key activities	• Calculate attrition rates by department, role, age, tenure, income, overtime, travel, and satisfaction. • Compare each rate with the 16.1% baseline and record group size (n). • Test for confounding, for example whether overtime overlaps with job level and income. • Identify high-risk segments and surprising results (for example, high performers leave at about the same rate as others).
Tools	Python, SQL, Power BI
Deliverables	Findings notes with candidate insights.
Exit criteria	Every insight has a rate, a group size, and a caveat.

Stage 6: Measures and validation

Objective	Implement KPIs once, correctly, and prove they match the source.
Key activities	• Write explicit DAX measures: Headcount, Attrition Count, Attrition Rate, Overall Attrition Rate, Rate vs Overall, Avg Monthly Income, Avg Tenure. • Store them in a dedicated measures table. • Reconcile totals: headcount 1,470, leavers 237, rate 16.1%. • Spot-check segment rates against the Python or SQL results.
Tools	Power BI (DAX)
Deliverables	Measure list and validation log.
Exit criteria	All reconciliations pass.

Stage 7: Visualisation and report design

Objective	Present the analysis so each page answers one business question.
Key activities	• Build Page 1 (where attrition concentrates), Page 2 (what drives it), and Page 3 (who is being lost and what could be saved). • Use takeaway-style titles, one accent colour for attrition, and reference lines for the overall rate. • Show group size in tooltips or labels. • Add slicers and check cross-filtering; do not rely on colour alone.
Tools	Power BI Desktop
Deliverables	HR attrition report (.pbix) and screenshots.
Exit criteria	Every business requirement maps to at least one visual.

Stage 8: Insights and communication

Objective	Turn results into findings a non-technical stakeholder can act on.
Key activities	• Write key findings, recommendations, and limitations in plain language. • State that correlation is not causation and that the data is synthetic. • Prepare a short summary slide or one-page brief.
Tools	Markdown, Word, PowerPoint
Deliverables	README insights section and summary.
Exit criteria	A non-technical reader can state the top three findings and next actions.

Stage 9: Review and acceptance

Objective	Confirm the solution meets the agreed requirements.
Key activities	• Test the report against each acceptance criterion in the BRD. • Build a traceability matrix from requirements to visuals. • Get feedback from a peer or mentor standing in for the sponsor. • Log feedback and changes made.
Tools	Word, Excel
Deliverables	Traceability matrix, review log, sign-off.
Exit criteria	All acceptance criteria met.

Stage 10: Deploy, monitor, and iterate

Objective	Make the work accessible and plan the next improvements.
Key activities	• Publish through Power BI Service, or share screenshots, a PDF, and the repository. • Document how a real refresh would work. • Log next steps: a predictive risk model (BR-09), a synthetic hire date for trend analysis, and cost-of-attrition estimates.
Tools	Power BI Service, GitHub
Deliverables	Publishing notes and roadmap.
Exit criteria	Repository is portfolio-ready and next steps are recorded.

4. Iteration loops

The stages are not strictly sequential. Expect these feedback loops and record at least one or two in the README:

- **Exploration back to cleaning (5 to 3):** a data quirk found during analysis, such as a role that spans departments, forces a change to preparation or the model.
- **Validation back to modelling (6 to 4):** a total that does not reconcile usually points to a relationship or grain problem.
- **Review back to design (9 to 7):** stakeholder feedback commonly changes layout, titles, or which breakdowns are shown.

5. Suggested repository structure

Folder	Contents
/docs	Business Requirements Document, life cycle document, data dictionary, traceability matrix, review log
/data/raw	Original dataset, unchanged
/data/clean	Cleaned dataset or dimension and fact tables, plus the cleaning log
/model	Model diagram, table definitions, measure list, validation log
/report	Power BI file (.pbix) and report screenshots
/insights	Exploratory findings, recommendations, and summary slide
README.md	Project overview, business questions, key findings, limitations, and how to reproduce

6. Risks to manage across the life cycle

- Synthetic, imbalanced data: state clearly that findings illustrate method, not real-world truth.
- Small segments: show group size beside every rate to avoid overinterpretation.

- Confounding: treat driver findings as associations, and present scenarios such as the overtime estimate as illustrations, not forecasts.
- No date field: trends are out of scope unless a synthetic hire date is added and labelled as constructed.